

In-Class

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!**

1. (20 pts) Determine the radius of convergence of each of the following series:

a.
$$\sum_{n=1}^{\infty} (-1)^n \frac{n+1}{n} (2z+i)^{2n}$$

b.
$$\sum_{n=1}^{\infty} (4-2i^n)^n (z-i)^n$$

2. (20 pts) Let G be a region in $\mathbb{C}$ and let $f \in \mathcal{A}(G)$, $f = u + iv$. Prove that if $u^2 = v^2$ on G , then f is constant on G .
3. (20 pts) Find all solutions of the equation $z^i = -2$.
4. (20 pts) Let $f(z) = e^{\frac{2+2z}{1-z}}$. Let T denote the unit circle, i.e., $T = \{z : |z| = 1\}$. Let γ be the subarc of T which goes, in the upper half-plane, from i to -1 . Identify and sketch the image of γ under f .
5. (10 pts) Let M be the Möbius transformation which maps $1, 0, i$ to $i, 1, -1$, resp. Find a formula for M .

6. (10 pts) The unit disk D , whose boundary is the circle C , which passes through the three points $1, i, -1$, is divided into four congruent subregions by the real and imaginary axes, say, D_1, D_2, D_3 and D_4 . (See figure to the right.) Let M be the Möbius transformation given by

$$M(z) = \frac{(2+i)z-i}{iz+i}.$$

Find/identify the images E_j of each subregion D_j under M , i.e., find $E_j = M(D_j)$, $j = 1, 2, 3, 4$.

